Number

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Estimation**

In a class of 30 students, 15 students play football and 20 students play basketball. Every student plays at least one of the two sports. Using a Venn diagram, find the number of students who play both football and basketball.

Let F be the set of students who play football and B be the set of students who play basketball. We are given that the total number of students is 30. The number of students who play football is $|F| = 15$. The number of students who play basketball is $|B| = 20$. Since every student plays at least one of the two sports, the union of the two sets is equal to the total number of students: $|F \cup B| = 30$. We can use the principle of inclusion-exclusion for two sets, which states.

2 **[Foundational]** • **Target: Rounding**

A number line is shown below. What is the value of the number indicated by the arrow?

The number line shows integers from -5 to 7. The arrow points directly to the mark labelled '0'. Therefore, the value of the number indicated by the arrow is 0. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

3 **[Foundational]** • **Target: Four operations with whole numbers**

The histogram shows the distribution of scores for a group of students in a mathematics test. The total number of students who scored between 0 and 40 is 32. The frequency density for the class interval $20 \leq x < 40$ is 0.8. Calculate the number of students who scored between 20 and 40.

The area of a bar in a histogram represents the frequency of the corresponding class interval. The area is calculated by frequency density multiplied by the class width. The class interval $20 \leq x < 40$ has a width of $40 - 20 = 20$. The frequency density for this interval is given as 0.8. Therefore, the frequency (number of students) in this interval is calculated as: Frequency = Frequency Density $\times$ Class Width Frequency = 0.8×20 Frequency = 16 Thus, 16 students scored between 20 and 40.

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4 [Foundational] • Target: Order of operations with whole numbers

The diagram shows a number line. The distance between the points marked A and B is 15 units. The coordinate of A is -7. Find the possible coordinates of B.

Let the coordinate of point B be x. The distance between two points on a number line is the absolute difference of their coordinates. Therefore, the distance between A and B is $|x-(-7)| = |x+7|$. We are given that this distance is 15 units. So, we have the equation $|x+7| = 15$. This gives us two possibilities: $x+7 = 15$ or $x+7 = -15$. Solving the first equation, $x = 15-7 = 8$. Solving the second equation, $x = -15-7 = -22$. Therefore, the possible coordinates of B are -22 and 8.

5 [Foundational] • Target: Using technology for operations with whole numbers

A bag contains n marbles, of which 4 are green. Two marbles are taken at random without replacement. If the probability that both are green is $\frac{2}{15}$, find the total number of marbles n.

$P(\text{both green}) = \frac{4}{n} \times \frac{3}{n-1} = \frac{12}{n(n-1)}$. Set equal to $\frac{2}{15}$: $\frac{12}{n(n-1)} = \frac{2}{15} \Rightarrow 2n(n-1) = 180 \Rightarrow n(n-1) = 90 \Rightarrow n^2 - n - 90 = 0 \Rightarrow (n-10)(n+9) = 0$. Since $n > 0$, $n = 10$. [M1 for setting up probability equation, M1 for quadratic n(n-1)=90, A1 for 10]

6 [Foundational] • Target: Check reasonableness of calculations

Set A has 12 elements and set B has 8 elements. If $A \cap B eq \varnothing$, what is the maximum possible number of elements in $A \cup B$?

Formula: $n(A \cup B) = n(A) + n(B) - n(A \cap B) = 12 + 8 - n(A \cap B) = 20 - n(A \cap B)$. To maximize the union, we must minimize the intersection. Since $A \cap B eq \varnothing$, the minimum possible value of $n(A \cap B)$ is 1. Thus maximum $n(A \cup B) = 20 - 1 = 19$. [M1 for maximizing union by minimizing intersection to 1, A1 for 19]

7 [Foundational] • Target: Estimation

The graph of the quadratic function $y = x^2 - 4x + 3$ is shown. The graph of the linear function $y = x - 1$ is also shown. The graphs intersect at points A and B. Find the coordinates of point A.

To find the intersection points, we set the two equations equal to each other: $x^2 - 4x + 3 = x - 1$. Rearranging the terms to form a quadratic equation: $x^2 - 5x + 4 = 0$. Factoring the quadratic equation: $(x-1)(x-4) = 0$. This gives us two possible x-values for the intersection points: $x = 1$ or $x = 4$. For $x = 1$, we can find the corresponding y-coordinate using the linear equation $y = x - 1$: $y = 1 - 1 = 0$. So, one intersection point is $(1, 0)$. For $x = 4$, we can find the corresponding y-coordinate.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Rounding

The diagram shows the graph of the quadratic function $f(x) = x^2 + ax + b$. The graph passes through the points $(-1, 8)$ and $(3, 0)$. Find the values of a and b.

The function is given by $f(x) = x^2 + ax + b$. Since the graph passes through $(-1, 8)$, we have $f(-1) = 8$. Substituting $x = -1$ into the function: $(-1)^2 + a(-1) + b = 8$ $1 - a + b = 8$ $-a + b = 7$ (*) Since the graph passes through $(3, 0)$, we have $f(3) = 0$. Substituting $x = 3$ into the function: $(3)^2 + a(3) + b = 0$ $9 + 3a + b = 0$ $3a + b = -9$ (**) Now we have a system of two linear equations with two variables: 1) $-a + b = 7$ 2) $3a + b = -9$ Subtract equation (1) from equation (2).

9 [Intermediate] • Target: Four operations with whole numbers

In the right-angled triangle ABC, $\angle ABC = 90^\circ$. If $AB = 5$ cm and $BC = 12$ cm, find the length of the hypotenuse AC.

The triangle ABC is a right-angled triangle with $\angle ABC = 90^\circ$. We are given the lengths of the two shorter sides, AB = 5 cm and BC = 12 cm. We can use the Pythagorean theorem to find the length of the hypotenuse AC. The Pythagorean theorem states that in a right-angled triangle, the square of the hypotenuse (the side opposite the right angle) is equal to the sum of the squares of the other two sides. $AC^2 = AB^2 + BC^2$. Substituting the given values: $AC^2 = 5^2 + 12^2$. $AC^2 = 25 + 144$. $AC^2 = 169$.

10 [Intermediate] • Target: Order of operations with whole numbers

The diagram shows a right-angled triangle ABC, with the right angle at B. The length of side AB is 10 cm and the length of side BC is 24 cm. A semicircle is drawn with AC as its diameter. Calculate the area of the semicircle in terms of π.

First, we need to find the length of the hypotenuse AC using the Pythagorean theorem: $AC^2 = AB^2 + BC^2$. $AC^2 = 10^2 + 24^2 = 100 + 576 = 676$. So, $AC = \sqrt{676} = 26$ cm. The radius of the semicircle is half the diameter AC, so the radius $r = \frac{26}{2} = 13$ cm. The area of a semicircle is given by $\frac{1}{2}\pi r^2$. Substituting the radius, we get $\frac{1}{2}\pi(13)^2 = \frac{1}{2}\pi(169) = 84.5\pi$ cm^2. Common error: Using the sides AB or BC as diameter, or for.

11 [Intermediate] • Target: Using technology for operations with whole numbers

The diagram shows the graph of the quadratic function $y = x^2 + bx + c$. The graph intersects the x-axis at $(1, 0)$ and $(5, 0)$. Find the values of b and c.

Since the graph intersects the x-axis at $(1, 0)$ and $(5, 0)$, these are the roots of the quadratic equation $x^2 + bx + c = 0$. We can use Vieta's formulas, which state that for a quadratic equation $ax^2 + bx + c = 0$, the sum of the roots is $-b/a$ and the product of the roots is c/a. In this case, $a = 1$. Sum of roots: $1 + 5 = 6$. Therefore, $-b/1 = 6$, which means $b = -6$. Product of roots: $1 \times 5 = 5$. Therefore, $c/1 = 5$, which means $c = 5$. Thus, the values are $b = -6$ and $c = 5$.

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12 [Intermediate] • Target: Check reasonableness of calculations

In a right-angled triangle, one of the acute angles is 30° and the side opposite to this angle is 6 cm. Calculate the length of the hypotenuse.

Let the right-angled triangle be ABC, with the right angle at C. Let angle A be 30°. The side opposite to angle A is BC, which is given as 6 cm. The hypotenuse is AB. We use the sine function: $\sin(A) = \frac{\text{opposite}}{\text{hypotenuse}}$. So, $\sin(30^\circ) = \frac{BC}{AB}$. We know that $\sin(30^\circ) = \frac{1}{2}$. Therefore, $\frac{1}{2} = \frac{6}{AB}$. Solving for AB, we get $AB = 2 \times 6 = 12$ cm.

13 [Intermediate] • Target: Estimation

A rectangular prism has a length of 5 cm, a width of 3 cm, and a height of 4 cm. Calculate the volume of the prism.

The volume of a rectangular prism is calculated by the formula: Volume = length × width × height. Given: Length = 5 cm Width = 3 cm Height = 4 cm Volume = 5 cm × 3 cm × 4 cm Volume = 15 cm^2 × 4 cm Volume = 60 cm^3 Therefore, the volume of the prism is 60 cm^3.

14 [Intermediate] • Target: Rounding

The graph of the function $y = f(x)$ is shown below. Find the value of $f(0)$.

The graph represents the function $y = f(x)$. To find the value of $f(0)$, we need to find the y-coordinate of the point where the graph intersects the y-axis. The y-axis is where $x = 0$. Looking at the graph, when $x = 0$, the corresponding y-value is 3. Therefore, $f(0) = 3$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Four operations with whole numbers • Cross-Lesson: 6.01 Trigonometry

The diagram shows the graph of the quadratic function $y = x^2 - 4x + k$. The graph intersects the x-axis at point P(1, 0) and the y-axis at point Q. Determine the value of k.

The graph of the quadratic function $y = x^2 - 4x + k$ passes through the point P(1, 0). Substituting the coordinates of P into the equation, we get: $0 = (1)^2 - 4(1) + k$ $0 = 1 - 4 + k$ $0 = -3 + k$ $k = 3$. Therefore, the value of k is 3.

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16 [Challenging] • Target: Order of operations with whole numbers • Cross-Lesson: 7.01 Transformations & Vectors

In the diagram below, lines AB and DE are parallel. Points A, C, and D are collinear, and points B, C, and E are collinear. If $AC = 6$ cm, $CD = 9$ cm, and $DE = 12$ cm, find the length of AB.

Since AB is parallel to DE, we can identify two similar triangles: ΔABC and ΔDEC. This is due to the property of alternate interior angles being equal when lines are parallel, and vertically opposite angles at C being equal. Specifically, $\angle BAC = \angle CDE$ and $\angle ABC = \angle DEC$ (alternate interior angles). Also, $\angle ACB = \angle DCE$ (vertically opposite angles). Because the triangles are similar, the ratio of their corresponding sides is equal.

17 [Challenging] • Target: Using technology for operations with whole numbers • Cross-Lesson: 8.01 Probability & Statistics

A point $A(3, 5)$ is mapped to $A'(1, 3)$ by a single transformation. The same transformation is then applied to a second point $B(x, y)$, mapping it to $B'(x - 2, y - 2)$. The vector representing the displacement from A to A' is $\vec{v}$. A point $C(2, 1)$ is reflected in the line $y = x$ to give point C'. Calculate the magnitude of the vector $A\vec{C}'$.

The transformation that maps $A(3, 5)$ to $A'(1, 3)$ is a translation. The translation vector $\vec{v}$ can be found by subtracting the coordinates of A from A': $\vec{v} = A' - A = (1 - 3, 3 - 5) = (-2, -2)$. This means the translation moves each point 2 units to the left and 2 units down. The problem states that point $B(x, y)$ is mapped to $B'(x - 2, y - 2)$ by the same transformation, which confirms our translation vector is correct. Next, point $C(2, 1)$ is reflected in the line $y = x$. A reflection in the line.

18 [Challenging] • Target: Check reasonableness of calculations • Cross-Lesson: 9.01 Functions

A survey of 100 students about their preferred social media platforms yielded the following information: - 55 students use Platform A. - 62 students use Platform B. - 18 students use neither Platform A nor Platform B. - x students use both Platform A and Platform B. Find the value of x.

Let A be the set of students who use Platform A and B be the set of students who use Platform B. We are given the total number of students is 100. Number of students who use neither A nor B is 18. Therefore, the number of students who use at least one platform is $100 - 18 = 82$. This means $|A \cup B| = 82$. We are given $|A| = 55$ and $|B| = 62$. The formula for the union of two sets is $|A \cup B| = |A| + |B| - |A \cap B|$. Here, $|A \cap B|$ represents the number of students who use both platforms, which is giv.

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19 [Challenging] • Target: Estimation • Cross-Lesson: 10.01 Sets & Venn Diagrams

A triangle ABC has vertices A(2, 3), B(5, 1), and C(3, 5). This triangle is subjected to a sequence of transformations. First, it is reflected in the line $y = x$ to produce triangle A'B'C'. Then, triangle A'B'C' is translated by the vector $(-3; 1)$ to produce triangle A''B''C''. ; Determine the coordinates of the vertices of triangle A''B''C''.

Let the original triangle ABC have vertices A(-2, 2), B(1, 3), and C(-1, 1). Step 1: Reflection in the line $y = x$. When a point (x, y) is reflected in the line $y = x$, its image is (y, x). Applying this to triangle ABC: A(-2, 2) becomes A'(2, -2). B(1, 3) becomes B'(3, 1). C(-1, 1) becomes C'(1, -1). Step 2: Translation of triangle A'B'C'. Triangle A'B'C' is translated by the vector $(-3; 1)$. This means we subtract 3 from the x-coordinate and add 1 to the y-coordinate of each vertex. Applying th.

20 [Challenging] • Target: Rounding • Cross-Lesson: 2.01 Algebra & Graphs

The diagram shows a point A and its image A' after a single transformation. The coordinates of A are (2, 1) and the coordinates of A' are (5, 3). The transformation can be represented by the vector $(p; q)$. Find the value of p.

The transformation from point A to its image A' can be represented by a translation vector. The change in the x-coordinate is the difference between the x-coordinate of A' and the x-coordinate of A. The change in the y-coordinate is the difference between the y-coordinate of A' and the y-coordinate of A. The translation vector is given by $(x_{A'} - x_A; y_{A'} - y_A)$. In this case, $x_A = 2$, $y_A = 1$, $x_{A'} = 5$, and $y_{A'} = 3$. Therefore, the translation vector is.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Estimation

For the model $f(x) = 2x + 3$ in estimation, evaluate $f(2)$.

a. 7 ✓

b. 9

c. 4

d. 11

Substitute $x = 2$: $f(2) = 2(2) + 3 = 7$.

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22 [Foundational] • Target: Estimation

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

a. $x = 2$ and $x = 5$

b. $x = 2$ **and** $x = 4$ ✓

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

Factoring $(x - 2)(x - 4) = 0$ yields the roots $x = 2$ and $x = 4$.

23 [Foundational] • Target: Estimation

For the model $f(x) = 2x + 3$ in estimation, evaluate $f(5)$.

a. 13 ✓

b. 15

c. 10

d. 17

Substitute $x = 5$: $f(5) = 2(5) + 3 = 13$.

24 [Foundational] • Target: Rounding

For the model $f(x) = 3x + 4$ in rounding, evaluate $f(2)$.

a. 10 ✓

b. 13

c. 6

d. 14

Substitute $x = 2$: $f(2) = 3(2) + 4 = 10$.

25 [Foundational] • Target: Rounding

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

a. $x = 2$ and $x = 6$

b. $x = 2$ **and** $x = 5$ ✓

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

Factoring $(x - 2)(x - 5) = 0$ yields the roots $x = 2$ and $x = 5$.

26 [Foundational] • Target: Rounding

For the model $f(x) = 3x + 4$ in rounding, evaluate $f(5)$.

a. 19 ✓

b. 22

c. 15

d. 23

Substitute $x = 5$: $f(5) = 3(5) + 4 = 19$.

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27 [Foundational] • Target: Four operations with whole numbers

For the model $f(x) = 4x + 5$ in four operations with whole numbers, evaluate $f(2)$.

a. 13 ✓

b. 17

c. 8

d. 17

Substitute $x = 2$: $f(2) = 4(2) + 5 = 13$.

28 [Foundational] • Target: Four operations with whole numbers

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

a. $x = 2$ and $x = 7$

b. $x = 2$ **and** $x = 6$ ✓

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Factoring $(x - 2)(x - 6) = 0$ yields the roots $x = 2$ and $x = 6$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Estimation

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ **and** $x = 3$ ✓

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

Factoring $(x - 1)(x - 3) = 0$ yields the roots $x = 1$ and $x = 3$.

30 [Intermediate] • Target: Estimation

A data set recorded for estimation consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5 ✓

d. 18.5

The sum is $10 + 14 + 18 + 24 = 66$, and dividing by 4 gives 16.5.

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31 [Intermediate] • Target: Estimation

For the model $f(x) = 2x + 3$ in estimation, evaluate $f(4)$.

a. 11 ✓

b. 13

c. 8

d. 15

Substitute $x = 4$: $f(4) = 2(4) + 3 = 11$.

32 [Intermediate] • Target: Estimation

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ **and** $x = 6$ ✓

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

Factoring $(x - 4)(x - 6) = 0$ yields the roots $x = 4$ and $x = 6$.

33 [Intermediate] • Target: Rounding

A data set recorded for rounding consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5 ✓

d. 19.5

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

34 [Intermediate] • Target: Rounding

For the model $f(x) = 3x + 4$ in rounding, evaluate $f(3)$.

a. 13 ✓

b. 16

c. 9

d. 17

Substitute $x = 3$: $f(3) = 3(3) + 4 = 13$.

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35 [Intermediate] • Target: Rounding

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ **and** $x = 6$ ✓

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

Factoring $(x - 3)(x - 6) = 0$ yields the roots $x = 3$ and $x = 6$.

36 [Intermediate] • Target: Four operations with whole numbers

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ **and** $x = 5$ ✓

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

Factoring $(x - 1)(x - 5) = 0$ yields the roots $x = 1$ and $x = 5$.

37 [Intermediate] • Target: Four operations with whole numbers

For the model $f(x) = 4x + 5$ in four operations with whole numbers, evaluate $f(3)$.

a. 17 ✓

b. 21

c. 12

d. 21

Substitute $x = 3$: $f(3) = 4(3) + 5 = 17$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Four operations with whole numbers • Cross-Lesson: 9.01 Functions

For the model $f(x) = 4x + 5$ in four operations with whole numbers, evaluate $f(4)$.

a. 21 ✓

b. 25

c. 16

d. 25

Substitute $x = 4$: $f(4) = 4(4) + 5 = 21$.

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39 [Challenging] • Target: Four operations with whole numbers • Cross-Lesson: 10.01 Sets & Venn Diagrams

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ **and** $x = 8$ ✓

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

Factoring $(x - 4)(x - 8) = 0$ yields the roots $x = 4$ and $x = 8$.

40 [Challenging] • Target: Order of operations with whole numbers • Cross-Lesson: 2.01 Algebra & Graphs

A data set recorded for order of operations with whole numbers consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

41 [Challenging] • Target: Order of operations with whole numbers • Cross-Lesson: 3.01 Coordinate Geometry

For the model $f(x) = 5x + 6$ in order of operations with whole numbers, evaluate $f(4)$.

a. 26 ✓

b. 31

c. 20

d. 30

Substitute $x = 4$: $f(4) = 5(4) + 6 = 26$.

42 [Challenging] • Target: Order of operations with whole numbers • Cross-Lesson: 4.01 Mensuration

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ **and** $x = 8$ ✓

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

Factoring $(x - 3)(x - 8) = 0$ yields the roots $x = 3$ and $x = 8$.

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43 [Challenging] • Target: Using technology for operations with whole numbers • Cross-Lesson: 5.01 Geometry

A data set recorded for using technology for operations with whole numbers consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

44 [Challenging] • Target: Using technology for operations with whole numbers • Cross-Lesson: 6.01 Trigonometry

For the model $f(x) = 6x + 7$ in using technology for operations with whole numbers, evaluate $f(4)$.

a. 31 ✓

b. 37

c. 24

d. 35

Substitute $x = 4$: $f(4) = 6(4) + 7 = 31$.

45 [Challenging] • Target: Check reasonableness of calculations • Cross-Lesson: 7.01 Transformations & Vectors

For the model $f(x) = 2x + 8$ in check reasonableness of calculations, evaluate $f(3)$.

a. 14 ✓

b. 16

c. 6

d. 18

Substitute $x = 3$: $f(3) = 2(3) + 8 = 14$.

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